CalcsLive Agent

Design and review CalcsLive calculations and create persistent articles. Excel bridge features will appear automatically when available.

Reviewed CalcsLive Script

Summary: Calculation completed with 1 output

Hollow Cylinder Mass Calculation

	Kind	Description	Symbol	Expression	Value	Unit
0	Input	Material density	ρ		1,000.0000	kg/m³
1	Input	Outer radius of cylinder	R		0.5000	m
2	Input	Wall thickness	t		0.0100	m
3	Input	Cylinder height	h		1.0000	m
4	Output	Mass of the topless cylinder	m	ρ * pi * (R^2 - (R - t)^2) * h	31.1018	kg

Category Metadata

Create Calculation Article

Create Article

The last reviewed `run_calcslive_script` payload is ready for approval.

Article Title

Cylinder Mass Calculation

Article Description

Calculates the mass of a cylinder based on material density, radius, and height.


```
{
  "pqs": [
    {
      "sym": "\u03c1",
      "unit": "kg/m\u00b3",
      "description": "Material density",
      "value": 1000
    },
    {
      "sym": "R",
      "unit": "m",
      "description": "Outer radius of cylinder",
      "value": 0.5
    },
    {
      "sym": "t",
      "unit": "m",
      "description": "Wall thickness",
      "value": 0.01
```

```
    },
    {
      "sym": "h",
      "unit": "m",
      "description": "Cylinder height",
      "value": 1.0
    },
    {
      "sym": "m",
      "unit": "kg",
      "description": "Mass of the topless cylinder",
      "expression": "\u03c1 * pi * (R^2 - (R - t)^2) * h"
    }
  ],
  "title": "Cylinder Mass Calculation",
  "description": "Calculates the mass of a cylinder based on material density, radius, and height."
}
```


Hi! I can help you design a CalcsLive calculation, review it, persist it, and, when Excel is connected, bridge it to and from Excel with automatic updates.

please calculate mass of a cylinder.

The calculation for the mass of a cylinder ran successfully with your default inputs. Here's a summary:

- **Inputs Used:**
  - Material density (ρ): 1000 kg/m^3
  - Cylinder radius (r): 0.5 m
  - Cylinder height (h): 1.0 m
- **Output Result:**
  - Mass (m): 785.398 kg

This uses the formula: mass = density × π × radius^2 × height.

The script was reviewed and worked without issues. Would you like to:

- Change any inputs or units?
- Persist this as a CalcsLive article for future use?
- Load it into Excel for editing (if Excel is connected)?

If you provide more details, I can refine this!

please change to a topless cylinder with equal thickness.

The updated calculation for a topless (hollow) cylinder with wall thickness ran successfully. I changed the model to use an outer radius and thickness, resulting in a new formula for the mass.

- **Inputs Used:**
  - Material density (ρ): 1000 kg/m^3
  - Outer radius (R): 0.5 m
  - Wall thickness (t): 0.01 m

  - Cylinder height (h): 1.0 m
- **Output Result:**
  - Mass (m): 31.102 kg

This uses the formula: mass = density × π × (outer radius^2 - (outer radius - thickness)^2) × height.

The script worked without any issues. Would you like to:

- Adjust any inputs or units (e.g., change material or dimensions)?
- Persist this as a CalcsLive article?
- Load it into Excel for editing (if Excel is connected)?

Let me know how you'd like to proceed!

Ask me to 'Calculate the values and write them back to Excel'...